# Frequency response

Practice questions · Day 07 · Fri 30 Oct · pack D4

## 5. Practice

### From your course files

**T4 Q3 (tutorial).** The following input-output pairs have been observed during the operation of various LTI systems. Determine the frequency response of each system.

(a) $x(n) = \left(\frac{1}{2}\right)^n u(n) \xrightarrow{T_1} y(n) = \left(\frac{1}{8}\right)^n u(n)$

(b) $x(n) = e^{j\frac{\pi}{5}n} \xrightarrow{T_2} y(n) = 3e^{j\left(\frac{\pi}{5}n - \frac{\pi}{3}\right)}$

(c) $x(n) = u(n) - u(n-1) \xrightarrow{T_3} y(n) = \left(\frac{1}{5}\right)^{n+2} u(n+2)$

T4 Q4 ($y(n) = x(n) + x(n-4)$): compute and sketch the magnitude and phase response, then the responses to (i) $x(n) = \cos(\frac{\pi}{4}n) + 3\sin(\frac{\pi}{3}n + \frac{\pi}{10})$ and (ii) $x(n) = 10 + 5\cos(\frac{2\pi}{5}n + \frac{\pi}{2})$) is worked in full as Example 1, which reproduces the official solution. Cover it and redo it without looking.

**T4 Q5 (tutorial).** An LTI system cannot produce frequencies at its output that are different from those applied in its input. Thus, if a system creates "new" frequencies, it must be nonlinear and/or time varying. Determine the frequency content of the outputs of the following systems to the input signal

$$x(n) = A\cos\left(\frac{\pi}{4}n\right)$$

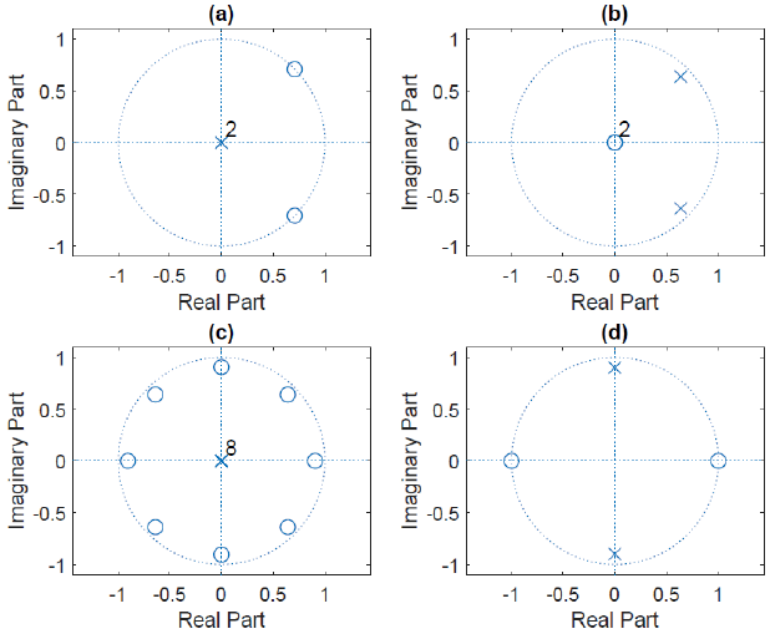
and conclude whether they are LTI.

(a) $y(n) = x(2n)$

(b) $y(n) = x^2(n)$

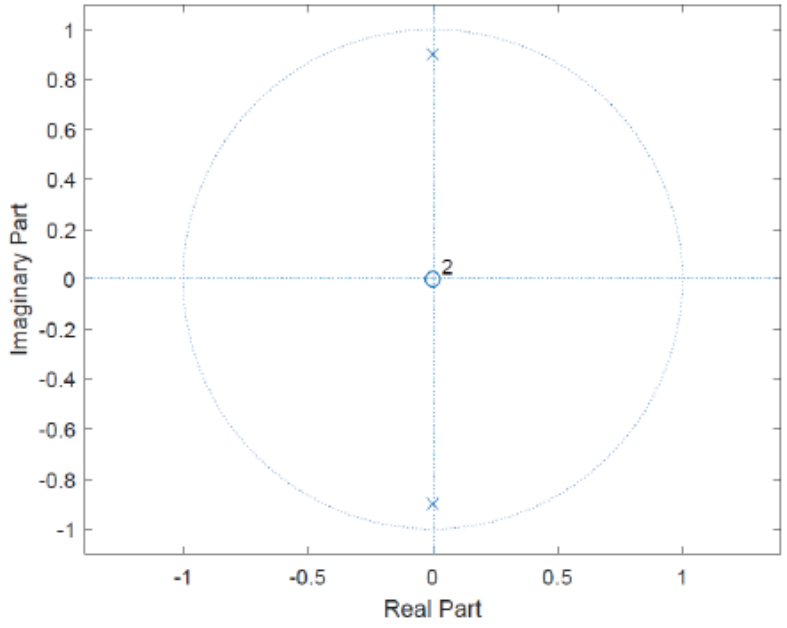
(c) $y(n) = x(n)\cos(\pi n)$

**T4 Q6 (tutorial).** Sketch roughly the magnitude of Fourier transforms, $|X(\omega)|$, corresponding to the pole-zero patterns given below.




T4 Q6 figure (from the tutorial sheet). o = zero, x = pole; the small numbers give multiplicity

**T4 Q7 (tutorial).** Sketch roughly the phase of the Fourier transform, $\theta(\omega)$, corresponding to the pole-zero pattern given below (assume $b_0 = 1$ and note that $N - M = 2$).




T4 Q7 figure (from the tutorial sheet)

## New questions

**N1.** [3] An FIR system has $h(n) = \{\underset{\uparrow}{1}, 2, 1\}$.

a. Find $H(\omega)$ in the form $e^{-j\alpha\omega}R(\omega)$ with $R(\omega)$ real. [1]
b. Find the output for $x(n) = 4\cos(\frac{\pi}{3}n + \frac{\pi}{4})$. [1]
c. Which input frequency in $[0, \pi]$ is completely blocked? [1]

**N2.** [4] A causal FIR system is known to have the form $y(n) = b_0x(n) + b_1x(n-1)$. Two experiments are observed: the constant input $x(n) = 1$ gives $y(n) = 3$, and $x(n) = (-1)^n$ gives $y(n) = -(-1)^n$.

a. Find $b_0$ and $b_1$. [2]
b. Find the output for $x(n) = \cos(\frac{\pi}{2}n)$. [2]

**N3.** [4] Apply $x(n) = A\cos(\frac{\pi}{3}n)$ to each system. Give the output frequencies in $[0, \pi]$ and state what you can conclude about LTI.

a. $y(n) = x(n)x(n-1)$ (b) $y(n) = x(n) + x(n-2)$ (c) $y(n) = x(-n)$ (d) $y(n) = 2x(n) + 1$

**N4.** [9] A notch filter is described by $y(n) = -0.81y(n-2) + x(n) + x(n-2)$.

a. Find $H(\omega)$. [1]
b. Find its poles and zeros and sketch the pole-zero plot. [2]
c. Using the vector method, evaluate $|H(\pi/4)|$ and $\theta(\pi/4)$. [3]
d. Find the steady-state output for $x(n) = 5 + 2\cos(\frac{\pi}{2}n) + 3\cos(\frac{\pi}{4}n + \frac{\pi}{3})$. [2]
e. Sketch $|H(\omega)|$ for $0 \le \omega \le \pi$. [1]

**N5.** [8] A first-order system is $H(\omega) = b_0 \dfrac{1 + e^{-j\omega}}{1 - 0.8e^{-j\omega}}$.

a. Choose $b_0 > 0$ so that $H(0) = 1$. [1]
b. Find $h(n)$, and confirm $\sum_n h(n) = H(0)$. [2]
c. Using the vectors from the zero and the pole to $e^{j\pi/2}$, find $|H(\pi/2)|$ and $\theta(\pi/2)$. [3]
d. Find the steady-state output for $x(n) = \cos(\frac{\pi}{2}n) + (-1)^n$. [1]
e. Find the 3 dB cutoff $\omega_c$, where $|H(\omega_c)| = 1/\sqrt{2}$. [1]

## 6. Quick recall check

1. What is the steady-state output of an LTI system to $x(n) = 7$ (a constant)?
2. $H(\omega_0) = -2$. Write the output for $\cos(\omega_0 n + 0.3)$.
3. A zero sits exactly on the unit circle at angle $2\pi/3$. What is $|H(2\pi/3)|$?
4. Does a pole at the origin change $|H(\omega)|$? Does it change $\theta(\omega)$?
5. An input at $\pi/5$ gives outputs at $\pi/5$ and $2\pi/5$. Could the system be LTI?